

Breast US: Guide for Beginners

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See the slide presentation [here](#).

US emerged in the 1950s and was initially used in breast imaging to distinguish cystic from solid breast lesions. As the technology evolved, it became possible to accurately characterize the risk of malignancy of solid breast masses based on shape, orientation, margin, echo patterns, posterior features, and vascularity. A standardized lexicon and classification system developed by the American College of Radiology known as the Breast Imaging Reporting and Data System (BI-RADS) improved reporting and tracking of sonographic findings. As US technology evolves, new techniques, including harmonics and Doppler US, may further improve evaluation of suspicious US findings.

US is an interactive and operator-dependent method. It allows real-time evaluation of breast pain or areas of palpable concern and direct interaction with the patient. US-guided percutaneous procedures are safe and reliable and allow precise needle visualization during the procedure.

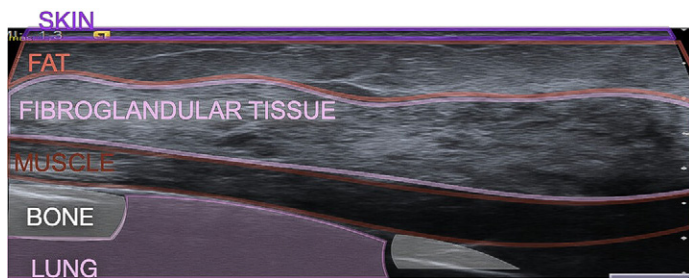


Figure 1. Gray-scale US image demonstrates the structures of the breast and chest wall.

US image quality depends on the operator's technique. Meticulous attention to detail is necessary to perform high-quality imaging. The area of interest should be located at the apex of the chest wall to allow the breast tissue to fall to the side, thinning the breast to facilitate imaging. The supine position is recommended for evaluation of medial breast findings, and the lateral decubitus position is recommended to better evaluate the lateral quadrants and axilla, especially in large breasts.

Evaluation of the breast must include all layers of tissue, including the skin, subcutaneous tissue, fibroglandular tissue, retromammary fat, and pectoral muscle (Fig 1). The depth must be adjusted to include the posterior pectoralis muscle and the focal zone in the anterior third between the skin and the thoracic wall.

Indications for breast US include evaluation of breast pain, masses, nipple discharge, and suspicious mammographic and MRI findings. US can guide percutaneous procedures, including biopsies, cyst aspiration, marker placement, preoperative localizations, and abscess drainage. Additionally, US may be used as supplementary screening in addition to mammography for high-risk patients who may not have access to MRI.

The accompanying slide presentation depicts the correct use of US techniques and features, such as high-frequency transducers, harmonics, and color Doppler, to avoid potential pitfalls (Fig 2).

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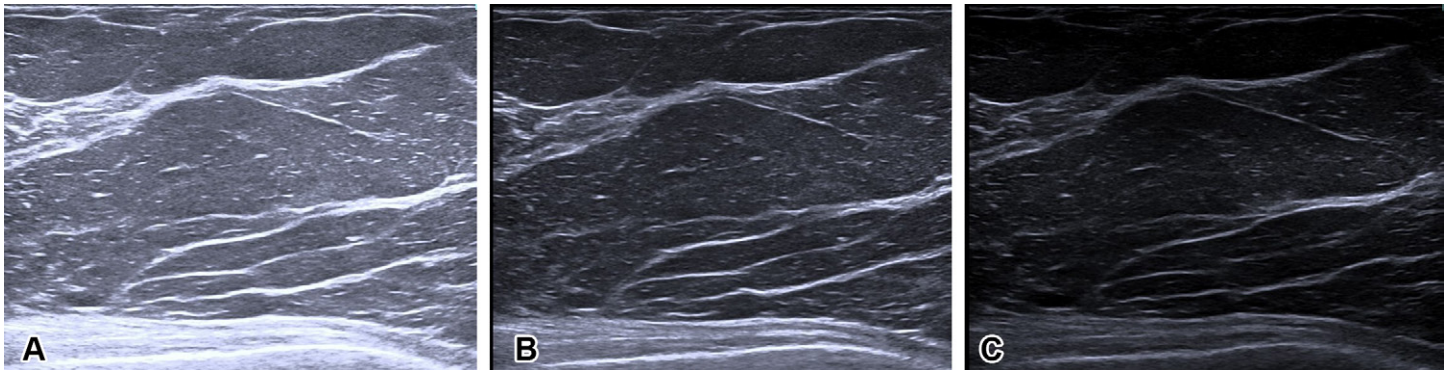


Figure 2. US images show normal breast tissue with different gains to demonstrate how gain should be adjusted to optimize breast US evaluation: normal breast tissue with excessive gain (A), adequate gain (B), and low gain (C).



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TEACHING POINTS

- US is widely available, does not involve ionizing radiation, does not require intravenous access for contrast material, and is a supplementary screening option to mammography for high-risk patients who may not have access to MRI, especially in patients with dense breast tissue.
- US is an interactive and operator-dependent method that allows real-time evaluation of suspicious findings and interaction with the patient.
- Correct use of US technique and features, such as harmonics and Doppler US, may help avoid potential pitfalls.

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